

# PIZZA SALES ANALYSIS USING SQL

- A DATA ANALYTICS PROJECT

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**SHODWE**  
Pizza Resto

# ABOUT PROJECT

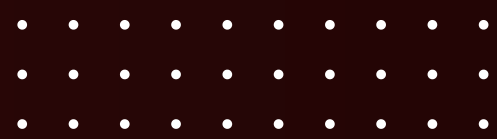
PIZZA SALES ANALYSIS

**This project analyzes pizza sales data using SQL to extract meaningful business insights. The goal is to understand customer ordering behavior, revenue trends, and product performance across different pizza categories and types. The analysis helps in making data-driven decisions to improve business performance.**

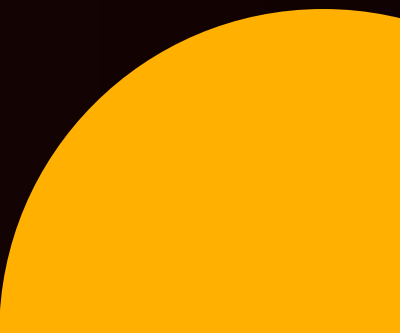




# OBJECTIVE



*THIS PROJECT AIMS TO ANALYZE PIZZA SALES DATA USING SQL TO UNDERSTAND OVERALL BUSINESS PERFORMANCE BY CALCULATING TOTAL ORDERS AND REVENUE, IDENTIFYING BEST-SELLING AND HIGHEST-PRICED PIZZAS, ANALYZING CATEGORY AND SIZE-WISE DISTRIBUTION, STUDYING TIME-BASED ORDER PATTERNS, AND DETERMINING TOP-PERFORMING PIZZAS BASED ON QUANTITY AND REVENUE ALONG WITH THEIR CONTRIBUTION AND TRENDS OVER TIME.*



# TOOLS USED



SQL (POSTGRESQL / MYSQL)

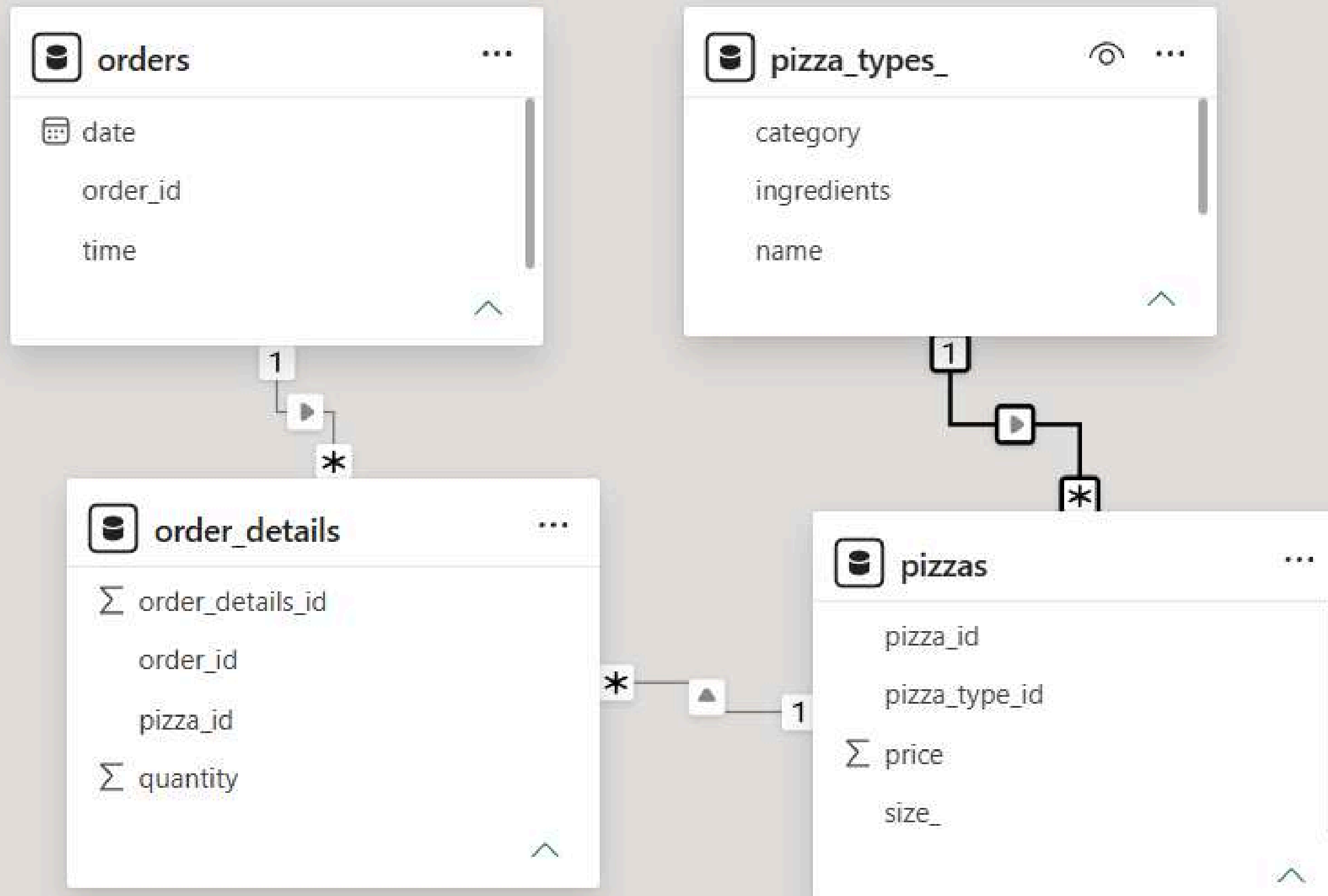
PG ADMIN

EXCEL (FOR VALIDATION)

POWER BI (FOR VISUALIZATION)



# schema design

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# RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

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```
SELECT  
    COUNT (ORDER_ID) AS NO_OF_ORDERS  
FROM  
    ORDERS;
```

|   | no_of_orders<br>bigint |
|---|------------------------|
| 1 | 21350                  |

# ***CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.***

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```
SELECT
    SUM(OD.QUANTITY * P.PRICE) AS TOTAL_REVENUE
FROM
    PIZZAS P
JOIN ORDER_DETAIL OD ON P.PIZZA_ID = OD.PIZZA_ID;
```

|   | total_revenue<br>numeric |
|---|--------------------------|
| 1 | 817860.05                |

# IDENTIFY THE HIGHEST-PRICED PIZZA.

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```
SELECT
  P.NAME,
  PI.PRICE
FROM
  PIZZA_TYPES_ P
  JOIN PIZZAS PI ON P.PIZZA_TYPE_ID = PI.PIZZA_TYPE_ID
ORDER BY
  PI.PRICE DESC
LIMIT
  1;
```

|   | name<br>character varying (50)  | price<br>numeric (5,2)  |
|---|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | The Greek Pizza                                                                                                      | 35.95                                                                                                        |



# IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

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```
select pi.size_ , sum(od.quantity) total_quantity from pizzas pi
join order_detail od
on pi.pizza_id = od.pizza_id
group by pi.size_
order by sum(od.quantity) desc
;
```

|   | size_<br>character varying (50) 🔒 | total_quantity<br>numeric 🔒 |
|---|-----------------------------------|-----------------------------|
| 1 | L                                 | 18956                       |
| 2 | M                                 | 15635                       |
| 3 | S                                 | 14403                       |
| 4 | XL                                | 552                         |
| 5 | XXL                               | 28                          |



# LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

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```
SELECT
    pt.name,
    SUM(od.quantity) AS total_quantity
FROM order_detail od
JOIN pizzas p
    ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

|   | name<br>character varying (50)  | total_quantity<br>numeric  |
|---|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1 | The Classic Deluxe Pizza                                                                                             | 2453                                                                                                            |
| 2 | The Barbecue Chicken Piz...                                                                                          | 2432                                                                                                            |
| 3 | The Hawaiian Pizza                                                                                                   | 2422                                                                                                            |
| 4 | The Pepperoni Pizza                                                                                                  | 2418                                                                                                            |
| 5 | The Thai Chicken Pizza                                                                                               | 2371                                                                                                            |

# JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

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```
SELECT
    PT.CATEGORY,
    SUM(OD.QUANTITY) AS TOTAL_QTY
FROM
    PIZZA_TYPES_ PT
    JOIN PIZZAS PI ON PI.PIZZA_TYPE_ID = PT.PIZZA_TYPE_ID
    JOIN ORDER_DETAIL OD ON PI.PIZZA_ID = OD.PIZZA_ID
GROUP BY
    PT.CATEGORY
ORDER BY
    SUM(OD.QUANTITY) DESC;
```

|   | category<br>character varying (50)  | total_qty<br>numeric  |
|---|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1 | Classic                                                                                                                  | 14888                                                                                                      |
| 2 | Supreme                                                                                                                  | 11987                                                                                                      |
| 3 | Veggie                                                                                                                   | 11649                                                                                                      |
| 4 | Chicken                                                                                                                  | 11050                                                                                                      |

# DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    EXTRACT(HOUR FROM o.time) AS hour_,
    COUNT(DISTINCT o.order_id) AS total_orders
FROM orders o
GROUP BY EXTRACT(HOUR FROM o.time)
ORDER BY hour_;
```

|    | hour_<br>numeric 🔒 | total_orders<br>bigint 🔒 |
|----|--------------------|--------------------------|
| 1  | 9                  | 1                        |
| 2  | 10                 | 8                        |
| 3  | 11                 | 1231                     |
| 4  | 12                 | 2520                     |
| 5  | 13                 | 2455                     |
| 6  | 14                 | 1472                     |
| 7  | 15                 | 1468                     |
| 8  | 16                 | 1920                     |
| 9  | 17                 | 2336                     |
| 10 | 18                 | 2399                     |
| 11 | 19                 | 2009                     |
| 12 | 20                 | 1642                     |
| 13 | 21                 | 1198                     |
| 14 | 22                 | 663                      |
| 15 | 23                 | 28                       |

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# GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

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```
WITH daily_sales AS (  
    SELECT  
        o.date,  
        SUM(od.quantity) AS pizzas_per_day  
    FROM orders o  
    JOIN order_detail od  
        ON o.order_id = od.order_id  
    GROUP BY o.date  
)  
SELECT  
    ROUND(AVG(pizzas_per_day), 2) AS avg_pizzas_per_day  
FROM daily_sales;
```

|   | avg_pizzas_per_day  |
|---|----------------------------------------------------------------------------------------------------------|
| 1 | 138.47                                                                                                   |

# -- DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
WITH total_revenue AS (  
    SELECT  
        pt.name AS pizza_type,  
        SUM(pi.price * od.quantity) AS total_revenue  
    FROM pizzas pi  
    JOIN pizza_types_ pt  
        ON pi.pizza_type_id = pt.pizza_type_id  
    JOIN order_detail od  
        ON pi.pizza_id = od.pizza_id  
    GROUP BY pt.name  
)  
SELECT *  
FROM total_revenue  
ORDER BY total_revenue DESC  
LIMIT 3;
```

|   | pizza_type<br>character varying (50) 🔒 | total_revenue<br>numeric 🔒 |
|---|----------------------------------------|----------------------------|
| 1 | The Thai Chicken Pizza                 | 43434.25                   |
| 2 | The Barbecue Chicken Piz...            | 42768.00                   |
| 3 | The California Chicken Piz...          | 41409.50                   |

# CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
WITH revenue_per_type AS (  
    SELECT  
        pt.name AS pizza_type,  
        SUM(pi.price * od.quantity) AS revenue  
    FROM pizzas pi  
    JOIN pizza_types pt  
        ON pi.pizza_type_id = pt.pizza_type_id  
    JOIN order_detail od  
        ON pi.pizza_id = od.pizza_id  
    GROUP BY pt.name  
)  
SELECT  
    pizza_type,  
    revenue,  
    ROUND(  
        100.0 * revenue / SUM(revenue) OVER(),  
        2  
    ) AS revenue_percentage  
FROM revenue_per_type  
ORDER BY revenue DESC;
```

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|----|--------------------------------------------|--------------------|-------------------------------|
|    | pizza_type<br>character varying (50)       | revenue<br>numeric | revenue_percentage<br>numeric |
| 1  | The Thai Chicken Pizza                     | 43434.25           | 5.31                          |
| 2  | The Barbecue Chicken Pizza                 | 42768.00           | 5.23                          |
| 3  | The California Chicken Pizza               | 41409.50           | 5.06                          |
| 4  | The Classic Deluxe Pizza                   | 38180.50           | 4.67                          |
| 5  | The Spicy Italian Pizza                    | 34831.25           | 4.26                          |
| 6  | The Southwest Chicken Pizza                | 34705.75           | 4.24                          |
| 7  | The Italian Supreme Pizza                  | 33476.75           | 4.09                          |
| 8  | The Hawaiian Pizza                         | 32273.25           | 3.95                          |
| 9  | The Four Cheese Pizza                      | 32265.70           | 3.95                          |
| 10 | The Sicilian Pizza                         | 30940.50           | 3.78                          |
| 11 | The Pepperoni Pizza                        | 30161.75           | 3.69                          |
| 12 | The Greek Pizza                            | 28454.10           | 3.48                          |
| 13 | The Mexicana Pizza                         | 26780.75           | 3.27                          |
| 14 | The Five Cheese Pizza                      | 26066.50           | 3.19                          |
| 15 | The Pepper Salami Pizza                    | 25529.00           | 3.12                          |
| 16 | The Italian Capocollo Pizza                | 25094.00           | 3.07                          |
| 17 | The Vegetables + Vegetables Pizza          | 24374.75           | 2.98                          |
| 18 | The Prosciutto and Arugula Pizza           | 24193.25           | 2.96                          |
| 19 | The Napolitana Pizza                       | 24087.00           | 2.95                          |
| 20 | The Spinach and Feta Pizza                 | 23271.25           | 2.85                          |
| 21 | The Big Meat Pizza                         | 22968.00           | 2.81                          |
| 22 | The Pepperoni, Mushroom, and Peppers Pi... | 18834.50           | 2.30                          |
| 23 | The Chicken Alfredo Pizza                  | 16900.25           | 2.07                          |
| 24 | The Chicken Pasta Pizza                    | 16701.75           | 2.04                          |

# ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
WITH daily_revenue AS (  
    SELECT  
        o.date,  
        SUM(pi.price * od.quantity) AS revenue  
    FROM orders o  
    JOIN order_detail od  
        ON o.order_id = od.order_id  
    JOIN pizzas pi  
        ON od.pizza_id = pi.pizza_id  
    GROUP BY o.date  
)  
SELECT  
    date,  
    revenue,  
    SUM(revenue) OVER (ORDER BY date) AS cumulative_revenue  
FROM daily_revenue  
ORDER BY date;
```

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|----|----------------------|-----------------------|-------------------------|
|    | date                 | revenue               | cumulative_revenue      |
| 1  | 2015-01-01           | 2713.85               | 2713.85                 |
| 2  | 2015-01-02           | 2731.90               | 5445.75                 |
| 3  | 2015-01-03           | 2662.40               | 8108.15                 |
| 4  | 2015-01-04           | 1755.45               | 9863.60                 |
| 5  | 2015-01-05           | 2065.95               | 11929.55                |
| 6  | 2015-01-06           | 2428.95               | 14358.50                |
| 7  | 2015-01-07           | 2202.20               | 16560.70                |
| 8  | 2015-01-08           | 2838.35               | 19399.05                |
| 9  | 2015-01-09           | 2127.35               | 21526.40                |
| 10 | 2015-01-10           | 2463.95               | 23990.35                |
| 11 | 2015-01-11           | 1872.30               | 25862.65                |
| 12 | 2015-01-12           | 1919.05               | 27781.70                |
| 13 | 2015-01-13           | 2049.60               | 29831.30                |
| 14 | 2015-01-14           | 2527.40               | 32358.70                |
| 15 | 2015-01-15           | 1984.80               | 34343.50                |
| 16 | 2015-01-16           | 2594.15               | 36937.65                |
| 17 | 2015-01-17           | 2064.10               | 39001.75                |

# DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

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```
WITH revenue_data AS (  
    SELECT  
        pt.category,  
        pt.name AS pizza_type,  
        SUM(pi.price * od.quantity) AS revenue  
    FROM pizzas pi  
    JOIN pizza_types pt  
        ON pi.pizza_type_id = pt.pizza_type_id  
    JOIN order_detail od  
        ON pi.pizza_id = od.pizza_id  
    GROUP BY pt.category, pt.name  
)  
ranked_data AS (  
    SELECT *,  
        RANK() OVER (  
            PARTITION BY category  
            ORDER BY revenue DESC  
        ) AS rnk  
    FROM revenue_data  
)  
SELECT  
    category,  
    pizza_type,  
    revenue  
FROM ranked_data  
WHERE rnk <= 3  
ORDER BY category, revenue DESC;
```

|    | category<br>character varying (50) 🔒 | pizza_type<br>character varying (50) 🔒 | revenue<br>numeric 🔒 |
|----|--------------------------------------|----------------------------------------|----------------------|
| 1  | Chicken                              | The Thai Chicken Pizza                 | 43434.25             |
| 2  | Chicken                              | The Barbecue Chicken Piz...            | 42768.00             |
| 3  | Chicken                              | The California Chicken Piz...          | 41409.50             |
| 4  | Classic                              | The Classic Deluxe Pizza               | 38180.50             |
| 5  | Classic                              | The Hawaiian Pizza                     | 32273.25             |
| 6  | Classic                              | The Pepperoni Pizza                    | 30161.75             |
| 7  | Supreme                              | The Spicy Italian Pizza                | 34831.25             |
| 8  | Supreme                              | The Italian Supreme Pizza              | 33476.75             |
| 9  | Supreme                              | The Sicilian Pizza                     | 30940.50             |
| 10 | Veggie                               | The Four Cheese Pizza                  | 32265.70             |
| 11 | Veggie                               | The Mexicana Pizza                     | 26780.75             |
| 12 | Veggie                               | The Five Cheese Pizza                  | 26066.50             |



**THANK YOU**

**FOR ATTENTION**

